

Markscheme

Mathematics: analysis and approaches

Practice question bank

33. (a)

Consider the function $f(x) = \frac{6x^2 - 6x + 4}{(x-1)^2(x^2+1)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+1}$.

attempt to write $A(x-1)(x^2+1) + B(x^2+1) + (Cx+D)(x-1)^2$ (M1)

$$6x^2 - 6x + 4 = A(x-1)(x^2+1) + B(x^2+1) + (Cx+D)(x-1)^2 \quad \text{A1}$$

$$\text{substituting } x = 1: 4 = 2B \Rightarrow B = 2 \quad \text{A1}$$

comparing coefficients of x^3 , x^2 , x^1 and the constant term (M1)

$$x^3: 0 = A + C \quad \text{A1}$$

$$\text{constant: } 4 = -A + B + D \quad \text{A1}$$

$$\text{solving simultaneously (using the } x^2 \text{ and } x^1 \text{ equations too): } A = 1, C = -1, D = 3 \quad \text{A1}$$

[7 marks]

(b)

Hence, by differentiating your partial fraction form term by term, find $f'(2)$.

differentiate each term (quotient rule on the last term) (M1)

$$f'(x) = -\frac{1}{(x-1)^2} - \frac{4}{(x-1)^3} + \frac{x^2 - 6x - 1}{(x^2+1)^2} \quad \text{A1}$$

$$f'(2) = -1 - 4 - \frac{9}{25} = -\frac{134}{25} = -5.36 \quad \text{A1}$$

[3 marks]